

Homework #5

Due: 4/3/26 by 5 PM

A web-server is managing customer purchase data for an online retailer, selling computer hard drives (somehow). The retailer has a limited supply of product, and therefore must implement a **Queue** in order to most effectively serve the customers. Your **Queue** will contain several **CustomerOrder** objects, which will contain the essential data a customer may enter to order something. Furthermore, your **Queue** must use a linked data structure, you're heavily encouraged to use **LinkedList** as a basis for this assignment. The code for **LinkedList** will be available for you on Brightspace

Note: You will lose considerable points if you fail to use a **Queue** and fail to use a Linked data structure. If you use an Array to do this, you're probably doing something wrong. There are no Tree's in this assignment, that will be later on. Furthermore, **you should not import Java's LinkedList, and will be penalized heavily if you do so.**

Prepare the following Class's:

- **CustomerOrder:**
 - Contains instance variables for name, date (use a string here), quantity
 - Note: Name and date will be unused, but should be included for completeness
 - Contains a default constructor to make a **null** customer
 - Contains a parameterized constructor to set up a customer
 - Implement a method to ship this customer a product. This will simply decrement quantity by one, **unless** quantity would become negative.

- **OrderQueue:**
 - Contains a **LinkedList** of **CustomerOrders**
 - Contains instance variables for stock (a number), size of queue
 - Contains a default constructor to set up an empty queue with no stock
 - Contains a parameterized constructor to set up an order queue, with a stock that is passed as a parameter.
 - Implement a method to add a new order to the queue. This method should construct a **CustomerOrder** (taking all the necessary data), and **enqueue** it to the order queue.
 - Implement a method to add additional stock. This simple method should take a single integer, and merely add it to the instance variable **stock**
 - Implement a method that will take the customer at the front of the queue, and fulfill one of their orders by decrementing their quantity by one, and the stock in the **OrderQueue** by one. If the customers quantity becomes 0, they should be **dequeued**.
 - Implement a method that will sell the remaining stock to each customer at the front of the queue until the stock is 0. You should use a **while** loop to accomplish this.
 - Implement a **toString** method that simply prints the quantity of the **CustomerOrder** at the front of the queue.

- **OrderTester:**
 - Contains a main function that will:
 - Instantiate an **OrderQueue** with a stock of 18.
 - Enqueue three customer orders with a positive balance, and quantities of 7, 7, and 8.
 - Print the **OrderQueue**
 - Clear the stock by filling as many orders as possible.
 - Print the **OrderQueue**
 - Add an additional 10 units to stock
 - Add three additional customer orders, with quantities of 3, 3, 5
 - Clear the stock by filling as many orders as possible.
 - Print the **OrderQueue**

Expected output:

5
4
5

When you're ready to submit your work, package your **.java files** (and not .class files) in a zip file using the following naming convention:

lastname-firstname-cos225-hw05.zip

Submit this .zip file containing each of the above .java files, as well as your github link, to Brightspace link for Homework #5.